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$$x(t) = X_m \cos(\omega t + \phi) \qquad (4.69)$$

where X_m is the maximum value of $x(t)$, ω the angular frequency and ϕ the phase constant.

Similarly, the electromagnetic energy of the *LC* system is given by

$$U = \frac{1}{2}Li^2 + \frac{1}{2}\left(\frac{1}{C}\right)q^2 = \text{constant} \qquad (4.70)$$

Differentiating U with respect to time, we get

$$\frac{dU}{dt} = \frac{1}{2}L\left(2i\frac{di}{dt}\right) + \frac{1}{2C}\left(2q\frac{dq}{dt}\right) = 0$$

$$\text{or} \quad L\frac{d^2q}{dt^2} + \frac{1}{C}q = 0 \qquad (4.71)$$

$$\text{since } i = \frac{dq}{dt} \text{ and } \frac{di}{dt} = \frac{d^2q}{dt^2}$$

The general solution of equation (4.71) is of the form

$$q(t) = Q_m \cos(\omega t + \phi) \qquad (4.72)$$

where Q_m is the maximum value of $q(t)$, ω the angular frequency and ϕ the phase constant.

Current in the LC circuit

The current flowing in the *LC* circuit is obtained by differentiating *q(t)* with respect to time.

$$i(t) = \frac{dq}{dt} = \frac{d}{dt}\left[Q_m \cos(\omega t + \phi)\right]$$

$$= -Q_m \omega \sin(\omega t + \phi) \qquad \text{since } I_m = Q_m\omega$$

$$\text{or} \quad i(t) = -I_m \sin(\omega t + \phi) \qquad (4.73)$$

The equation (4.73) clearly shows that current varies as a function of time *t*. In fact, it is a sinusoidally varying alternating current with angular frequency ω.

Angular frequency of LC oscillations

By differentiating equation (4.72) twice, we get

$$\frac{d^2q}{dt} = -Q_m\omega^2 \cos(\omega t + \phi) \qquad (4.74)$$

Substituting equations (4.72) and (4.74) in equation (4.71), we obtain

$$L\left[-Q_m\omega^2 \cos(\omega t + \phi)\right] + \frac{1}{C}Q_m \cos(\omega t + \phi) = 0$$

Rearranging the terms, the angular frequency of *LC* oscillations is given by

$$\omega = \frac{1}{\sqrt{LC}} \qquad (4.75)$$

This equation is the same as that obtained from qualitative analogy.

Oscillations of electrical and magnetic energy

The electrical energy of the LC oscillator is

$$U_E = \frac{q^2}{2C} = \frac{Q_m^2}{2C}\cos^2(\omega t + \phi) \qquad (4.76)$$

The magnetic energy is

$$U_B = \frac{1}{2}Li^2 = \frac{Q_m^2}{2C}\sin^2(\omega t + \phi) \qquad (4.77)$$

If the two energies are plotted with an assumption of $\phi = 0$, we obtain Figure 4.58.

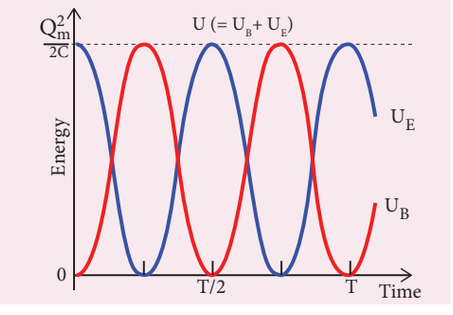


Figure 4.58 The variation of U_E and U_B as a function of time

From the graph, it can be noted that

(i) At any instant $U_E + U_B = \frac{Q_m^2}{2C}$ = constant

(ii) The maximum values of U_E and U_B are both $\frac{Q_m^2}{2C}$.

(iii) When U_E is Maximum, U_B is zero and vice versa.

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